100 DAYS QUALITY MANAGEMENT CHALLENGE

DRDO

DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION

INTEGRATED DEFENCE QUALITY ENGINEERING

Research Quality • Experimental Testing • Reliability Assessment
Failure Investigation • System Validation

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INDEPENDENT EDUCATIONAL CASE STUDY

A practical learning document connecting quality engineering principles with a hypothetical system-testing problem.
Not an official DRDO publication.

TECHNICAL LEARNING • EVIDENCE-BASED REASONING • ENGINEERING JUDGEMENT

HOW TO USE THIS REPORT

This report separates public organisational context, general engineering principles and hypothetical case-study details.

- **Organisational context:** Refer to DRDO's official public website for general information.
- **Engineering principles:** Technical explanations are general educational material, not a description of DRDO's internal methods.
- **Hypothetical details:** The scenario is invented for learning and does not represent a real DRDO incident or test.
- **Decision discipline:** A suspected cause is not a confirmed root cause; completing an action does not prove effectiveness.

Section	Purpose
1. Defence R&D; quality context	Quality, verification, validation and reliability.
2. Hypothetical problem	Define a test-performance variation without assuming its cause.
3. Critical quality characteristics	Requirements, evidence and acceptance logic.
4. Tools and analysis	Method selection, interpretation and limitations.
5. Root-cause investigation	Compare plausible explanations with evidence.
6. Risk assessment	Identify consequences and mitigation.
7. Engineering solution	Match action to supported cause.
8. Validation	Review action effectiveness and requirements evidence.
9. Interview preparation	25 original research-oriented questions with answers.
10. References and disclaimer	Traceable further reading and clear scope.

Core principle: sound engineering decisions require defined requirements, trustworthy measurements, documented evidence and appropriate technical judgement.

1. DEFENCE R&D; QUALITY CONTEXT

Defence research and development applies scientific and engineering methods to develop, evaluate and improve technologies. Quality engineering supports clear requirements, meaningful experiments, trustworthy measurements, risk awareness and evidence-based conclusions. This report discusses general principles; it does not describe DRDO's internal procedures.

Research-stage quality and production-stage quality

At the research stage, teams may be evaluating a concept, learning how factors affect performance or determining whether a design meets a defined need. Test methods, assumptions and conditions require careful control. Production-stage quality also emphasises repeatable manufacture, supplier controls and consistent conformity across units. These activities overlap, but the engineering questions are not identical.

Term	Meaning	Illustrative example
Quality Assurance (QA)	Planned activities intended to provide confidence that requirements will be met.	Reviewing whether the test plan, roles and records are defined.
Quality Control (QC)	Activities used to assess whether an output meets requirements.	Checking a test result against a predefined criterion.
Verification	Evidence that specified requirements have been fulfilled.	Checking a measured parameter against its stated requirement.
Validation	Evidence that a system is suitable for its intended use and context.	Evaluating whether demonstrated behaviour meets intended needs.
Reliability	Probability of performing a required function under stated conditions for a stated time.	Assessing failure-free operation over a defined period.
Maintainability	Ability to be retained in or restored to a specified condition under stated conditions.	Evaluating repair time using a defined method.
Repeatability	Closeness of results under the same defined conditions.	Repeated runs with the same setup and method.
Measurement uncertainty	A parameter describing dispersion of values reasonably attributable to the measurand.	Reporting a result with a justified uncertainty statement.

Why one successful test is not enough

A successful run shows that the observed criterion was met in that run under its recorded conditions. It does not by itself establish repeatability, performance throughout an operating range, long-term reliability or suitability for every intended use. Additional evidence depends on the requirement, risk, test design and acceptance rules.

Good records identify what was tested, under which conditions, with which equipment and configuration, how the result was evaluated and whether deviations occurred.

2. HYPOTHETICAL ENGINEERING PROBLEM

Scenario status: Entirely hypothetical. It is not an actual DRDO project, failure incident, internal procedure or test result.

Scenario: An engineering team is evaluating a hypothetical electronic subsystem under specified environmental conditions. During repeated tests, one measured performance parameter varies. Some runs satisfy the predefined acceptance criterion, while others approach or exceed an illustrative limit. The team must determine whether the variation is associated with the subsystem, environmental conditions, the test setup, instrumentation, execution differences or interactions between factors.

Problem statement

Determine whether the variation is real and technically meaningful, identify contributing causes using suitable evidence, and decide what action is justified. Do not label the subsystem defective before the evidence supports that conclusion.

Step	Key question	Evidence to preserve
1. Define requirement	What parameter, conditions and acceptance rule apply?	Approved requirement, method and revision.
2. Confirm data	Are values, units, timestamps and records complete?	Raw data, calculations, logs and processing steps.
3. Confirm setup	Was the intended configuration used?	Setup checklist, configuration records and deviations.
4. Review environment	Were conditions within the specified range?	Environmental readings and test logs.
5. Evaluate measurement	Is the method suitable and traceable?	Instrument identity, calibration status and uncertainty information.
6. Compare runs	Does variation associate with a condition or factor?	Stratified data, repeats and experimental design.
7. Decide action	What conclusion is justified by evidence?	Reviewed findings, limitations and authorised decision record.

Do not move acceptance limits after seeing the results merely to make a test pass. If a requirement is unclear or unsuitable, review it through the applicable authorised process before making a conformity decision.

3. CRITICAL QUALITY CHARACTERISTICS

A critical quality characteristic is a measurable or assessable property important to meeting a requirement, managing risk or demonstrating suitability. Characteristics and limits must come from applicable engineering requirements—not from this illustrative report.

Characteristic	Why it matters	Possible evaluation	Evidence
Performance	Shows whether required function is achieved.	Compare measured response with predefined criterion.	Requirement, method, raw result and evaluation.
Environmental behaviour	Conditions may affect response.	Test under specified environmental conditions.	Condition logs, setup and deviations.
Repeatability	Shows consistency under defined conditions.	Repeat controlled runs.	Run-by-run results and conditions.
Measurement suitability	Poor methods can mislead decisions.	Assess range, resolution, calibration and uncertainty.	Instrument and method evaluation.
Reliability	Addresses performance over a defined period.	Use suitable model and evidence plan.	Failure data, exposure and assumptions.
Configuration control	Different configurations may yield incomparable results.	Check item and setup identification.	Configuration identifiers and change records.
Safety and risk	Consequences may be significant.	Apply the relevant risk method.	Hazard analysis, controls and review.
Traceability	Supports review and reconstruction.	Link item, setup, method and result.	Identifiers, records and approvals.

Requirements before results

Before testing, identify the parameter, units, conditions, measurement method, acceptance criteria, handling of invalid runs and required records. Resolve ambiguity through the authorised process before deciding conformity.

4. QUALITY TOOLS AND TECHNICAL ANALYSIS

Choose a tool to answer a defined engineering question. A tool output is evidence to interpret—not automatic proof of a cause or a guarantee that a system is acceptable.

A. Structured test planning

Defines objectives, requirements, controlled variables, conditions, methods, criteria and records.

When to use: Use before a test when consistent execution and reviewable evidence matter.

Limitation: A plan improves consistency but cannot guarantee representativeness of every real condition.

B. Check sheets and stratification

Organises data by run, condition, time, equipment, configuration or other relevant factors.

When to use: Use to see whether variation clusters around a condition or change.

Limitation: Patterns guide investigation; they do not prove causation.

C. Measurement-system evaluation

Reviews calibration status, resolution, repeatability, reproducibility and uncertainty.

When to use: Use when readings disagree or results are close to a limit.

Limitation: Calibration alone does not prove fitness for a particular task.

D. Design of Experiments (DOE)

Plans combinations of factor settings to estimate effects and interactions.

When to use: Use when several plausible factors may contribute and controlled testing is feasible.

Limitation: Conclusions depend on design, factor ranges, data quality and assumptions.

E. FMEA

Identifies failure modes, effects, causes and controls to support risk prioritisation.

When to use: Use prospectively to organise risks and preventive actions.

Limitation: Scores are aids; a single score cannot replace judgement.

F. Root-cause analysis

Uses methods such as Five Whys, fishbone diagrams or fault trees to examine hypotheses.

When to use: Use after defining the problem and assembling reliable evidence.

Limitation: A plausible explanation is not proof; test the proposed cause.

G. Statistical analysis

Uses summaries, plots, control charts or models appropriate to the data.

When to use: Use when sampling and data structure support the method.

Limitation: Small or dependent samples and changing conditions can mislead.

H. Reliability engineering

Evaluates probability of required function over stated time and conditions.

When to use: Use to plan reliability evidence and interpret failure/exposure data.

Limitation: Claims depend on test design, failure definition and assumptions.

I. Verification and validation

Verification checks requirements; validation evaluates intended-use suitability.

When to use: Use at appropriate development and evaluation stages.

Limitation: A narrow verification test does not prove suitability for every use.

J. Corrective-action review

Links containment, cause confirmation, action and effectiveness monitoring.

When to use: Use after a problem is understood and action is required.

Limitation: Action completion is not proof of effectiveness.

Illustrative DOE example

Suppose engineers want to study whether two controlled factors—A (low/high) and B (low/high)—are associated with a measured response. A planned 2 × 2 experiment compares all four combinations. Replication helps estimate variation; randomisation can reduce run-order bias; an interaction term tests whether the effect of A changes with B. This is a generic example, not a real DRDO test.

Factor A	Factor B	Record
Low	Low	Response, conditions and run details.
Low	High	Response, conditions and run details.
High	Low	Response, conditions and run details.
High	High	Response, conditions and run details.

Do not invent results for this table. It illustrates experimental structure only.

Useful statistical concepts

Mean: $x\blacksquare = (\Sigma x\blacksquare)/n$. It summarises the centre of observations but can hide spread or unusual values.

Sample standard deviation: $s = \sqrt{[\Sigma(x\blacksquare - x\blacksquare)^2/(n - 1)]}$. It describes sample spread; interpretation depends on data quality and sampling assumptions.

Uncertainty: A reported result may include an uncertainty statement such as $y \pm U$, but the meaning and coverage basis of U must be defined and justified.

Control charts are useful only when the data structure and sampling plan support them. Control limits describe statistical behaviour; they are not engineering specification limits. Cp and Cpk should not be applied automatically to small experimental datasets.

Reliability: careful interpretation

Reliability is the probability of performing a required function without failure for a stated time under stated conditions. MTBF, where suitable for a repairable-system model, is a statistical average under that model—not a promise that each unit will last that long. A small failure-free test does not establish near-perfect reliability.

For independent representative trials under a suitable Bernoulli model, zero failures in a small sample still leaves uncertainty about the population success probability. Reliability demonstration requires a defined objective, test conditions, sample or exposure plan and decision rule.

5. ROOT-CAUSE INVESTIGATION

The objective is to discriminate between plausible explanations rather than select the first explanation that seems reasonable. These are hypotheses to investigate, not confirmed causes.

Potential cause	Evidence	Method	Possible follow-up
Test setup variation	Setup records, connection checks, equipment status.	Setup audit; controlled repeat.	Standardise setup and verify procedure.
Environmental variation	Environmental readings and test timing.	Compare by condition; review sensor records.	Control or document relevant conditions.
Measurement limitation	Instrument ID, calibration, resolution, uncertainty.	Method review; suitable comparison where authorised.	Improve method or quantify limitation.
Configuration difference	Item/setup identifiers and change records.	Configuration comparison.	Restore intended configuration; assess affected results.
Execution difference	Procedure steps, deviations and logs.	Record review and procedure walk-through.	Clarify and control execution.
Subsystem behaviour	Repeatable data under verified conditions.	Controlled experiment and engineering analysis.	Investigate design/component if supported.

Potential cause	Evidence	Method	Possible follow-up
Factor interaction	Data across planned factor combinations.	DOE or other suitable analysis.	Confirm with follow-up evidence.

Investigation sequence

Problem definition → Requirement review → Data verification → Test-system evaluation → Hypothesis development → Controlled investigation → Evidence review → Root-cause confirmation

Keep these distinctions clear:

- **Observed variation:** what the records show.
- **Suspected cause:** a possible explanation that still needs testing.
- **Verified root cause:** a cause supported by adequate evidence for the decision.
- **Corrective action:** a change intended to address the confirmed cause.
- **Effectiveness evidence:** results showing whether the action achieved its intended outcome.

6. RISK ASSESSMENT AND MITIGATION

Risk-based thinking asks what could go wrong, how serious the consequences may be, what controls exist and what additional evidence or mitigation is required. Use the method appropriate to the project and its authorised requirements.

Risk	Potential consequence	Possible mitigation	Evidence
Invalid test setup	Misleading pass/fail conclusion.	Setup verification and controlled execution.	Checklist and repeatability evidence.
Unsuitable measurement	Incorrect decision near a limit.	Assess method and uncertainty.	Instrument/method review.
Uncontrolled configuration	Results may not be comparable.	Identify and control changes.	Configuration records and impact assessment.
Unverified cause	Ineffective action or recurrence.	Test hypotheses before acting.	Investigation and confirmatory evidence.
Insufficient reliability evidence	Overconfident reliability claim.	Define objective, exposure and decision rule.	Plan and statistical analysis.
Incomplete records	Decision cannot be reconstructed.	Record data, conditions, deviations and approvals.	Traceability review.

Risk workflow

Identify risk → Analyse risk → Evaluate against criteria → Select treatment → Monitor and review

A risk matrix or FMEA score is a prioritisation aid, not an objective truth. Consider severity, likelihood, detectability, uncertainty and consequences according to the applicable method. This is not DRDO's official risk matrix.

7. PROPOSED ENGINEERING SOLUTION

The appropriate action depends on evidence. Hardware redesign should not be the default if the issue is actually measurement-related, configuration-related or caused by uncontrolled test conditions.

Evidence indicates	Possible response	Verification
Test conditions uncontrolled	Improve setup controls and document conditions.	Repeat under controlled conditions; review records.
Measurement method unsuitable	Assess instrument, method, uncertainty and range.	Show method supports the intended decision.
Configuration differs	Confirm intended configuration and review affected results.	Verify configuration identity and repeat relevant tests.
Procedure execution varies	Clarify steps, roles and deviation handling.	Observe execution and review run records.
Subsystem behaviour supported	Conduct authorised design/component investigation.	Use defined requirements and verification plan.
Factors may interact	Use a planned experiment or suitable analysis.	Confirm with follow-up evidence.

Corrective-action record

Stage	Question to answer
Problem	What was observed, against which requirement?
Evidence	Are raw data and conditions reliable and complete?
Cause	What evidence supports the cause?
Action	Does the action address the cause and consequences?
Verification	Was the action implemented as intended?
Effectiveness	Do suitable follow-up results show control of the problem?

Any release, retest acceptance, design change or risk-acceptance decision must follow applicable requirements and authorised procedures. This educational report cannot authorise such a decision.

8. VALIDATION AND EXPECTED IMPACT

Plan validation and effectiveness review before declaring success. Evidence must match the requirement and the risk of an incorrect conclusion.

Check	Review question	Evidence
Requirements	Are criteria current and unambiguous?	Approved requirement and method.
Test setup	Was the intended setup verified?	Checklist, configuration and deviations.
Measurement	Can the method support the decision?	Calibration status, uncertainty and suitability review.
Execution	Were tests performed as planned?	Run logs and documented deviations.
Results	Do results satisfy the predefined rule?	Raw data, calculations and reviewed interpretation.
Repeatability	Is behaviour sufficiently consistent for the question?	Appropriate repeated runs and conditions.
Corrective action	Was the action implemented and did it work?	Implementation and effectiveness evidence.
Residual risk	Are remaining risks understood?	Risk review and authorised decision record.

Do not confuse these outcomes:

- **Action completed:** the task was performed.
- **Implementation verified:** evidence confirms the action was put in place.
- **Effectiveness demonstrated:** suitable follow-up evidence supports the intended effect.
- **Requirement demonstrated:** valid evidence supports the required system behaviour under defined conditions.

Potential benefits include more trustworthy test evidence, better understanding of variation, fewer unsupported conclusions, improved traceability and more defensible decisions. These are potential benefits, not measured outcomes from an actual DRDO programme.

9. TECHNICAL INTERVIEW PREPARATION

25 independently developed questions for research and engineering learning. They are not represented as actual DRDO interview questions.

CATEGORY A — RESEARCH AND EXPERIMENTAL QUALITY

Q1. Why should a test objective be defined before selecting the experimental procedure?

Answer: The objective determines what to measure, which conditions matter, what evidence is needed and how results will be judged. Without it, a test may generate data that do not answer the engineering question.

Q2. What is the difference between repeatability and reproducibility in experimental testing?

Answer: Repeatability concerns closeness under the same defined conditions. Reproducibility concerns agreement when specified conditions change, such as equipment, operator or laboratory, depending on the study.

Q3. Why distinguish exploratory experiments from formal acceptance tests?

Answer: Exploratory tests help develop understanding and hypotheses. Formal acceptance tests evaluate predefined criteria using an authorised method; exploratory evidence does not automatically replace them.

Q4. What should a test protocol contain?

Answer: Objectives, item/configuration, conditions, equipment, method, data to record, acceptance criteria, deviation handling and required approvals.

Q5. Why are experimental records important for independent review?

Answer: They let reviewers reconstruct what was tested, under which conditions, how data were handled and why the conclusion was reached.

CATEGORY B — TEST DATA AND MEASUREMENT

Q6. How can measurement uncertainty affect a decision near an acceptance limit?

Answer: Uncertainty describes dispersion reasonably associated with the result. Near a limit it may affect the conformity decision, so the applicable decision rule should state how uncertainty is handled.

Q7. Why does a calibrated instrument still need a suitability assessment?

Answer: Calibration establishes a relationship with reference values under stated conditions. It does not alone prove suitable resolution, range, uncertainty or fitness for the specific decision.

Q8. Difference between measurement error and uncertainty?

Answer: Error is the difference between a measured value and a reference or true value where available. Uncertainty characterises dispersion of values reasonably attributable to the measurand. They are related but not interchangeable.

Q9. Why should anomalous results not be automatically deleted?

Answer: An anomaly may reflect a real failure, unusual condition, data error or measurement issue. Removing it without a justified predefined rule can bias analysis and hide evidence.

Q10. How can engineers separate system variation from test-setup variation?

Answer: Review records, evaluate the measurement method, stratify by conditions and run controlled repeats. Use a designed experiment if several factors must be separated.

CATEGORY C — EXPERIMENTAL DESIGN AND FAILURE INVESTIGATION

Q11. How does Design of Experiments help identify factors associated with variation?

Answer: DOE plans factor combinations so effects can be estimated more efficiently than informal testing, provided the design and assumptions suit the question.

Q12. Why can factor interactions matter?

Answer: An interaction means the effect of one factor depends on another factor's level. Ignoring it can produce misleading conclusions or an ineffective solution.

Q13. What is the role of replication?

Answer: Replication provides information about variability and can improve precision when analysed appropriately. It is not the same as repeated measurements of one sample without accounting for dependence.

Q14. Why test a root-cause hypothesis before corrective action?

Answer: A plausible explanation may be wrong. Testing it reduces the risk of treating a symptom instead of the cause.

Q15. When might fault-tree analysis be more suitable than Five Whys?

Answer: Fault-tree analysis structures how combinations of events can lead to a top-level undesired event. Five Whys is simpler for tracing a focused causal chain.

CATEGORY D — RELIABILITY AND SYSTEM EVALUATION

Q16. Why is one successful test insufficient to establish long-term reliability?

Answer: It gives limited evidence under one set of conditions. Long-term claims need a defined period, failure definition, representative conditions and adequate evidence.

Q17. What is reliability demonstration testing for?

Answer: It evaluates whether evidence supports a stated reliability objective under defined assumptions, conditions and a decision rule.

Q18. How should a failure-free test with a small sample be interpreted?

Answer: State that no failures were observed in that sample and those conditions. Do not conclude the population never fails; uncertainty and confidence bounds matter.

Q19. Difference between reliability and availability?

Answer: Reliability concerns performing without failure over a stated period. Availability concerns being operational when needed and may depend on reliability, repair time, maintenance and logistics.

Q20. Why specify conditions and time in reliability claims?

Answer: Failure behaviour depends on environment, workload, duty cycle, maintenance and time. A claim without these qualifiers is incomplete.

CATEGORY E — ADVANCED SCENARIO-BASED ENGINEERING JUDGEMENT

Q21. A system passes once but fails in a repeated run under apparently similar conditions. What should be examined first?

Answer: Preserve both results; confirm the requirement and test validity; compare configuration, conditions, setup, measurement method and execution records. Do not dismiss either run prematurely.

Q22. Two instruments give different readings for the same item. How should the discrepancy be investigated?

Answer: Check instrument identity, calibration, range, resolution, method, setup and environment. Review uncertainty and compare using a suitable reference or method where authorised.

Q23. A corrective action improves the average result but increases run-to-run variation. What evidence is needed?

Answer: Review requirements and risk, examine the spread and conditions, assess instability or interactions, and conduct appropriate follow-up tests. A better mean alone may not establish acceptability.

Q24. A reliability test has no failures but very few tested units. What can be concluded?

Answer: Only that no failures were observed in that sample under recorded conditions. High population reliability is not established without considering sample size, exposure, assumptions and confidence bounds.

Q25. A result meets the numerical limit but the approved test procedure was not followed. Can it automatically be accepted?

Answer: No. The deviation may affect validity or comparability. Document and assess it under the authorised process; review or retesting may be required.

Interview principle: Explain the evidence, assumptions and limits of a conclusion—not only the name of a tool.

10. REFERENCES AND DISCLAIMER

Organisational reference

DRDO official website: https://www.drdo.gov.in/ — use for general organisational information and publicly available material. It does not verify the hypothetical scenario in this report.

Technical references for further study

- Montgomery, Douglas C. *Design and Analysis of Experiments*. Wiley. Verify the edition used.
- Montgomery, Douglas C. *Introduction to Statistical Quality Control*. Wiley. Verify the edition used.
- O'Connor, Patrick D. T., and Andre Kleyner. *Practical Reliability Engineering*. Wiley. Verify the edition used.
- JCGM 100:2008. *Evaluation of measurement data — Guide to the expression of uncertainty in measurement (GUM)*. Joint Committee for Guides in Metrology.
- ISO 9001, *Quality management systems — Requirements*. Consult the applicable verified edition and applicability.
- ISO 31000, *Risk management — Guidelines*. Consult the applicable verified edition and context.
- ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*. Consult the applicable verified edition where relevant.

These are potentially relevant learning references, not a claim that every listed standard is mandatory or applicable to every DRDO programme. Verify edition and project-specific applicability before use.

Disclaimer

This document is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. It is not an official DRDO publication and is not endorsed by DRDO. The engineering scenario and any illustrative data are hypothetical and do not describe an actual DRDO project, incident, internal procedure or test result. Technical concepts are discussed for educational purposes. Actual engineering decisions must follow applicable requirements, approved procedures, safety obligations and authorised technical documentation.

FINAL TAKEAWAY: Quality Engineering in research is not simply about obtaining a passing result. It is about defining the question, designing a suitable test, trusting the measurement, investigating variation, managing risk and supporting conclusions with defensible evidence.